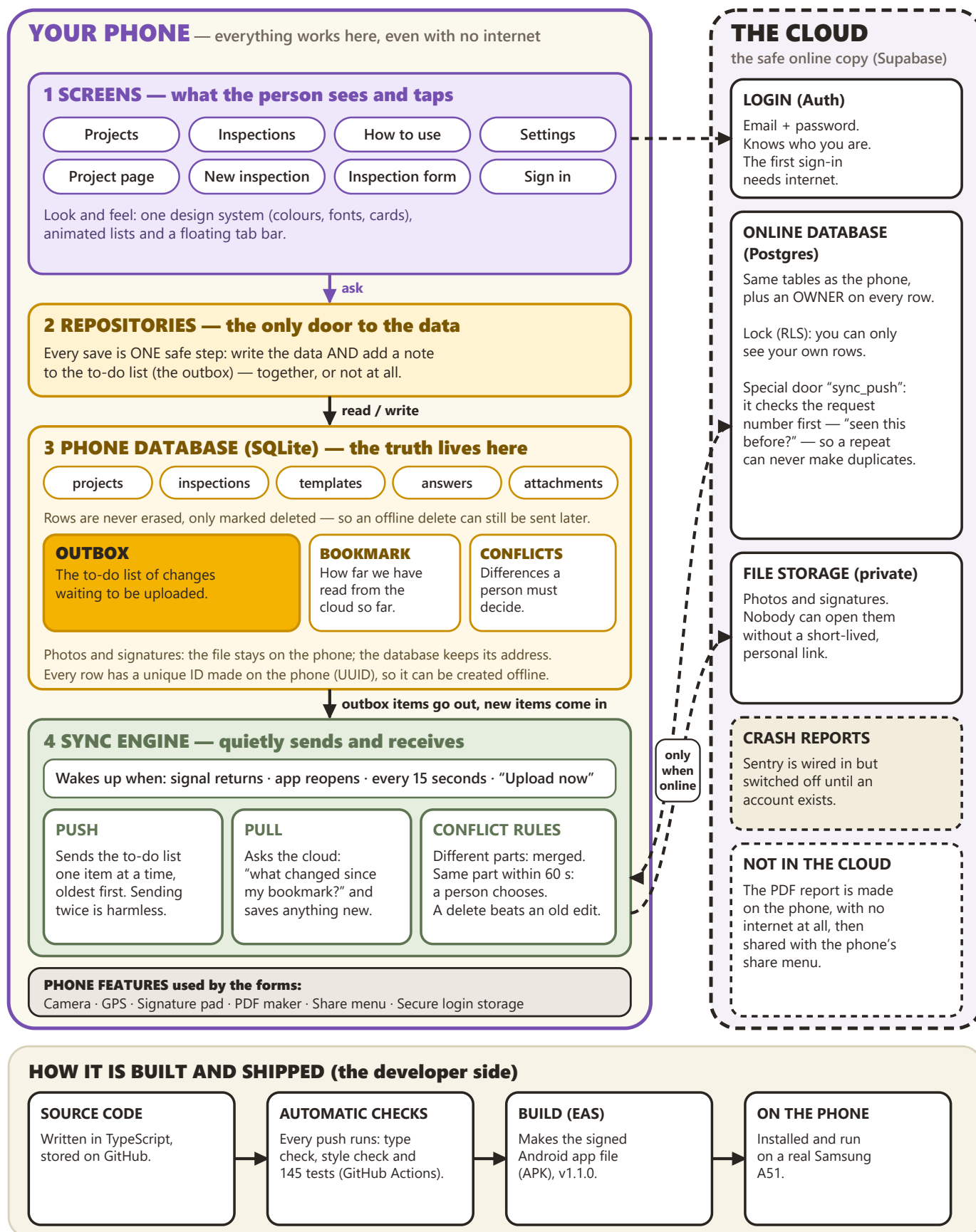


Fieldnote — architecture at a glance

The big picture: what lives on the phone, what lives in the cloud, and how the app is built and shipped. Read it top to bottom.



What happens when you save something

Follow the numbers 1 to 9. Steps 1–4 always work. Steps 5–9 happen by themselves when there is internet.

A • SAVING ON THE PHONE — always works, even with no signal

1 You type or tap

An answer, a photo, a new project...

2 The screen asks a repository

Screens never touch the database directly.

3 One safe step

Save the data AND add a note to the outbox — together, or not at all.

4 You see “Saved”

Instantly. Nothing waits for the internet.

B • SENDING TO THE CLOUD — only when internet is there

5 The engine wakes up

Signal returns, app reopens, 15-second check or “Upload now”.

6 PUSH one item at a time

Oldest first (a project before its inspections). Each item carries its own request number.

7 The cloud checks

“Seen this request before?” “Who is signed in?” Then it saves the item under your name.

8 The phone ticks it off

Item marked uploaded and removed from the outbox. Photos: the file is sent as well.

9 PULL what is new

Asks “what changed since my bookmark?”. Conflict rules decide what wins.

WHAT IF SOMETHING GOES WRONG?

No signal

Step 6 just waits. Nothing is lost; the to-do list keeps growing safely.

App closed halfway

The item stays in the outbox and is tried again. Sending twice is harmless.

Sent twice by mistake

The cloud recognises the request number and ignores the second copy.

Refused again and again

It is set aside (“dead-letter”). Only the Retry button tries it again.

Words used in the diagrams

Word	What it means
Outbox	A to-do list on the phone of every change that still has to be uploaded.
Sync	Quietly matching the phone and the cloud so both end up with the same information.
Push / Pull	Push = send my changes up. Pull = fetch other changes down.
Bookmark (cursor)	A note of how far the phone has already read from the cloud.
Safe to repeat (idempotent)	Sending the same request twice has the same result as sending it once — so retries never create duplicates.
Conflict	The same thing was changed in two places. Different parts are merged; the same part is decided by a person.
RLS (row-level security)	A lock in the online database: each signed-in person can only see and change their own rows.
Soft delete	A deleted item is only marked as deleted, so the delete can still be sent to the cloud.
UUID	A long unique ID made on the phone, so nothing needs the internet to get an ID.
Dead-letter	An item the cloud keeps refusing; it is set aside until a person taps Retry.